Simulating optimality on disk search algorithms with variable request spread

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We are trying to answer the question, “At what point/in what scenario are certain heuristics better/worse in terms of arm movements and fairness than an optimal algorithm for disk search?” We find this project interesting because we want to figure out if the spread of requests is different (eg. More requests on the ends, less in the middle, or the opposite) does that change which greedy heuristic is most optimal and which ensures the most fairness. This project will teach us not only more about each heuristic/new heuristics but also the optimal algorithm for disk search. It will also require us to make two algorithms to simulate disk searches and a visual representation of that happening. The challenges we expect to face include creating a program that keeps track of virtual time and implementing the visual representation of the program running (neither of us has taken graphics).

Rubric:

Choosing heuristics and finding an optimal disk seek algorithm to compare them to (optimal also can be expensive; this is just the comparison point).
{15%}

Producing a random request algorithm where you can choose a spread type, for example 30% chance a request is on track 1, 1% track 2, all the way to track 100, etc. {25%}. This would produce an array of objects where each object is a request and the time it would be submitted to the disk.

Producing a running program where you can choose between optimal or one of the heuristics to simulate a disk search. The program will take the result of our random request algorithm and produce an array depicting the arm movements over a disk search and/or an array depicting when each request was finished.
{25%}

Visual representation of the program running {15%}

Produce final visuals showing optimality and fairness for each heuristic with each request spread type. These visuals should be products of aggregates of several runs of each algorithm under a circumstance.
{20%}